

ATHIRA KRISHNAN R.

Ph.D. Research Scholar · Robotics & Embodied AI

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Dear Hiring Manager,

I am writing to express my strong interest in a **Robotics Research position at Tata Consultancy Services Research & Innovation (TCS R&I)**. Having previously served as a System Engineer at TCS, where I worked with Rapid Labs and Smart Mobility. I have spent my doctoral journey at the IIT Hyderabad under guidance of Prof Sumohana S Chappayya and Prof K Sri Rama Murty. My career represents a continuous path of innovation from developing autonomous parking solutions at TCS to pioneering AI solutions robotic systems.

My doctoral research, titled *"Multimodal Embodied Intelligence for Mobile Robots: From Perception to Navigation and Semantic Mapping,"* focuses on developing intelligent robotic systems capable of perceiving, reasoning, and autonomously navigating complex environments. My research spans multimodal perception, embodied AI, semantic mapping, visual-language navigation, robot learning, and autonomous navigation, with an emphasis on building complete end-to-end systems that can transition from simulation to deployment.

A cornerstone of my Ph.D. has been leading the vision and perception research for a **DRDO-sponsored Vision project** focused on autonomous mobile robots. As the technical lead for vision modules, I directed the development of critical capabilities including semantic segmentation, monocular depth estimation, and scene understanding. Beyond algorithm design, I specialized in the rigorous optimization of deep learning models for real-time inference on resource-constrained edge computing platforms. My research further explores the frontier of Multimodal Embodied Intelligence, integrating visual, auditory, and semantic data to enable robots to reason within their environments. I have developed learning-based navigation frameworks and semantic mapping pipelines that allow agents to process diverse sensory inputs for more effective goal-directed navigation. This work has direct real-world applicability in sectors ranging from industrial automation and healthcare to disaster response, where robots must operate autonomously in unstructured, dynamic settings.

My research has resulted in publications at leading international venues, including **IEEE ICC, TMLR, SPCOM, NCC, and PREMI**, reflecting my commitment to advancing the state of the art in embodied AI and robotics. Beyond research publications, I actively contribute to the robotics community through professional activities and technical collaborations, and I enjoy translating innovative research ideas into deployable intelligent systems.

My earlier experience at Tata Consultancy Services provided me with a strong foundation in software engineering, collaborative development, and large-scale system design. It taught me the importance of building reliable, scalable solutions while working within multidisciplinary teams. Returning to TCS in a research capacity would allow me to combine these engineering principles with my research experience to contribute to technologies that have both scientific significance and industrial impact.

I am particularly drawn to TCS Research & Innovation because of its long-term vision for advancing Artificial Intelligence, Robotics, and Physical AI through interdisciplinary research. I would welcome the opportunity to contribute to research in embodied intelligence, autonomous robots, multimodal perception, robot learning, semantic understanding, and intelligent autonomous systems while collaborating with researchers across diverse domains to solve challenging real-world problems.

I believe my combination of industrial software engineering experience, research leadership in a DRDO-sponsored robotics project, hands-on development of autonomous robotic systems, and a strong publication record uniquely positions me to contribute to TCS R&I. I would be delighted to discuss how my experience and research interests align with your ongoing initiatives.

Complementing my technical research is a deep commitment to professional leadership and community engagement. Serving as the Treasurer and GSAC Advisor for the **IEEE RAS Kerala Chapter** for over five years, I have spearheaded global-level initiatives and facilitated numerous technical workshops on ROS and AI. These roles have sharpened my ability to mentor multidisciplinary teams and communicate complex technical concepts to diverse audiences, ensuring that innovative research translates into community-wide progress.

Thank you for your time and consideration. I look forward to the opportunity to speak with you.

Sincerely,

Athira Krishnan R.

*Ph.D. Research Scholar, Department of Artificial Intelligence
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